

Ikkyum Kim

Integrated M.S. & Ph.D. Student,

Department of Electrical and Electronic Engineering,

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Contact Information

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Education

- ***Integrated M.S. & Ph.D.***, Dept. of Electrical and Electronic Engineering, Ulsan National Institute of Science and Technology (UNIST) (2024.9 – Present)
 - Advisor: Prof. Heechun Park
- ***B.S.***, Dept. of Electrical Engineering, Kookmin University (2018.3 – 2024.8)

Research Experiences

- ***Visiting Researcher***, Dept. of Electrical and Computer Engineering, University of Southern California (USC), Los Angeles, CA, USA, under Prof. Sung Kyu Lim (2025.10)
- ***Visiting Researcher***, Semiconductor Design Automation (SeDA) Lab, Dept. of Electrical and Electronic Engineering, Ulsan National Institute of Science and Technology (UNIST) (2024.3 – 2024.8)

Research Interests

- Machine learning for electronic design automation (ML-EDA)
 - Graph neural networks for physical design optimization
 - Design space exploration with learning-based search
- Clock-aware design methodologies
 - Clock tree / clock distribution network synthesis
 - Clock–signal co-optimization
- Backside design methodologies for advanced technology nodes
 - Backside clock distribution network (BS-CDN) synthesis
 - Backside power delivery network (BS-PDN) aware physical design
- Physical design methodologies for 2.5D and 3D integration
- Emerging computing architectures
 - Technology mapping for memristive logic-in-memory

Publications (accepted and published only)

International Conferences (9)

- ***Ikkyum Kim*** and Heechun Park, “DiffTherm: A Differentiable Thermal Formulation for Cell-Via Co-Optimization in 3D ICs”, IEEE/ACM Asia and South Pacific Design Automation Conference (ASP-DAC), 2027
- Geonhyeong Park, ***Ikkyum Kim***, and Heechun Park, “A Comprehensive Clock Tree Synthesis with Useful Skew for 3D ICs”, IEEE/ACM Asia and South Pacific Design Automation Conference (ASP-DAC), 2027

- Gangmin Jeon, **Ikkyum Kim**, Junghyun Yoon, and Heechun Park, “Thermal-Aware Chiplet Placement and Shape Co-Optimization under D2D Interconnect Constraints for 2.5D Integration”, IEEE/ACM Asia and South Pacific Design Automation Conference (ASP-DAC), 2027
- **Ikkyum Kim**, Junghyun Yoon, Gyumin Kim, Jooyeon Jeong, and Heechun Park, “Comprehensive BS-CDN Synthesis Framework with GNN Clustering and MILP-Based Optimization”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), 2026
- **Ikkyum Kim**, Geonhyeong Park, and Heechun Park, “BS-CSO: Backside-Assisted Clock-Signal Co-Optimization Framework”, IEEE/ACM International Symposium on Low Power Electronics and Design (ISLPED), 2026
- Junghyun Yoon, **Ikkyum Kim**, Gyumin Kim, Sojung Park, and Heechun Park, “Standard Cell Layout Generation: Methodological Evolution and Architectural Impacts”, IEEE/ACM Asia and South Pacific Design Automation Conference (ASP-DAC), 2026
- Yehyun Shin[†], **Ikkyum Kim**[†], Minho Park[†], Junghyun Yoon, Seunghun Baek, Seongmin Eum, Heesoo Yang, Yurim Choi, Jaeyong Jeong, Sanghyeon Kim, Haksoon Jung, Seongju Kim, Heechun Park, and Jimin Kwon, “Active BSCDN Benchmark Framework with Backside-Compatible CNFET Logic Technology”, IEEE International Electron Devices Meeting (IEDM), 2025 ([†]*Co-first authors*)
- **Ikkyum Kim** and Heechun Park, “Memristive Logic-in-Memory Implementation with Area Efficiency and Parallelism”, IEEE International Conference on Computer Design (ICCD), 2024
- **Ikkyum Kim** and Heechun Park, “3T1R Memristor Crossbar Architecture for Diverse Logic Implementation”, IEEE International SoC Design Conference (ISOC), 2024

International Journals (1)

- **Ikkyum Kim**, Minhong Kim, and Heechun Park, “A Parallelism-Driven, Area-Aware Technology Mapping Framework for Memristive Logic-in-Memory”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Early Access, 2026 (DOI: 10.1109/TCAD.2026.3699619)

Domestic Conferences (2)

- Minhong Kim, **Ikkyum Kim**, and Heechun Park, “Logic Partitioning Methodology for Utilizing Active Backside Layers”, Korean Conference on Semiconductors (KCS), 2026
- **Ikkyum Kim** and Heechun Park, “Logic-in-Memory Technology Mapping Framework for Memristor Crossbar with Maximized Parallelism”, Korean Conference on Semiconductors (KCS), 2024

Patents

- 권지민, 박희천, 박민호, 신예현, **김익결**, “클록 분배망의 독립적 배선 구조를 이용한 클록 신호 및 로직 신호의 완전 분리형 반도체 장치 및 시스템”, Korean Patent, Application No. 10-2025-019445, Filed December 9, 2025
- 모현선, 김대정, **김익결**, “난수 생성 장치”, Korean Patent, Application No. 10-2023-0065523, Filed May 22, 2023

Technical Skills

- **Commercial EDA**: Synopsys Design Compiler, PrimeTime, PrimeLib; Cadence Genus, Innovus, Tempus, Voltus
- **Open-Source and Research Tools**: OpenROAD, Yosys/ABC, Gurobi, PACT, HotSpot, COMSOL
- **Programming and ML**: PyTorch, scikit-learn; NumPy, SciPy, Numba; Python, Tcl, C++, Verilog

(Last updated: October 1, 2026)